GECON 7201
Applied Econometrics

Assignment 1
Due Thursday October 8, 2026

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Instructions

- Edit this Quarto document for your assignment. All answers must be typed in this document. Answers are to be written in the callout blocks (see below)
- Provide a link to your website in Nexus, so that I can view your assignment.
- Upload both a PDF and Quarto file to the nexus assignment portal.
- The use of AI is strictly prohibited for this assignment.
- Show your work
- Type your mathematical answers using LaTeX
- For questions marked R, include both your code and its output.
- You may use `solve()`, `t()`, `%*%`, `diag()`, `qr()`, `eigen()`, and other base R functions freely unless a question says otherwise.
- Please round numerical answers to two decimal places unless stated otherwise.

```
set.seed(3201)
```

Questions

Q1

Create a personal Quarto website hosted on GitHup pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

a. An "Index" or "Home" page. This page should include the following: - Your name - Your photo (optional) - Your credentials - An "About Me" section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

b. A blog page. This will be the landing site for future assignments [5 Marks]
c. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
d. Navigation bar with "Home" and "Blog" buttons. The navigation bar is edited in the "_quarto.yml" file. [2 Marks]

https://abaya-m.github.io/REPOSITORY-NAME/Meagan-s-Personal-Webite-2026/

Q2

Consider

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -3 \\ 0 & -3 & 6 \end{bmatrix}.$$

a. State the dimensions of A and identify a_{23}. [2 Marks]

```
A <- matrix(c(2,0,1,0,3,-1,1,-1,4), nrow = 3, byrow = TRUE)
dim(A)
A[2,3]
```

b. Is A a square matrix? Is it symmetric? Justify your answer using the definition $A = A^\top$, not just by inspection. [2 Marks]

A is a square Matrix because $A^T = A$, it is symmetric too because the elements are identical along the diagonal line.

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}, A^T = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

c. Is A a diagonal matrix? Explain why or why not. [1 Mark]

A is not a diagonal matrix. A diagonal matrix only has non-zero values along its main diagonal and zero in all elements not in the diagonal. This matrix has non-zero values in the elements that do not lie in the main diagonal

d. In R, Enter A and B into R. Use `isSymmetric()` to confirm your answer to part (b) for both matrices. [4 Marks]

```r
A <- matrix(c(2,0,1,0,3,-1,1,-1,4), nrow = 3, byrow = TRUE)
B <- matrix(c(1,2,0,2,5,-3,0,-3,6), nrow = 3, byrow = TRUE)
isSymmetric(A)
isSymmetric(B)
```

Q3

Using the same A and B from Question 1:

a. Compute $A + B$ and $3A - 2B$ by hand. [4 marks]

$$A + B \begin{bmatrix} 2+1 & 0+2 & 1+0 \\ 0+2 & 3+5 & -1-3 \\ 1+0 & -1-3 & 4+6 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 8 & -4 \\ 1 & -4 & 10 \end{bmatrix}$$

$$3A = \begin{bmatrix} 3*2 & 3*0 & 3*1 \\ 3*0 & 3*3 & 3*-1 \\ 3*1 & 3*-1 & 3*4 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 3 \\ 0 & 9 & -3 \\ 3 & -3 & 12 \end{bmatrix}$$

$$2B = \begin{bmatrix} 2*1 & 2*2 & 0 \\ 2*2 & 2*5 & 2*-3 \\ 0 & 2*-3 & 2*6 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 0 \\ 4 & 10 & -6 \\ 0 & -6 & 12 \end{bmatrix}$$

$$3A - 3B = \begin{bmatrix} 6-2 & 0-4 & 3-0 \\ 0-4 & 9-10 & -3-(-6) \\ 3-0 & -3-(-6) & 12-12 \end{bmatrix} = \begin{bmatrix} 4 & -4 & 3 \\ -4 & -1 & 3 \\ 3 & 3 & 0 \end{bmatrix}$$

b. Compute AB by hand, showing at least one entry's calculation in full using the summation formula $(AB)_{ij} = \sum_{\ell} a_{i\ell} b_{\ell j}$. [2 marks]

$$A*B = \begin{bmatrix} 2*1+0*2+0*1 & 2*2+0*5+1*-3 & 2*0+0*-3+1*6 \\ 0*1+3*2+-1*0 & 0*2+3*5+-1*-3 & -1*0+-3*3+-1*6 \\ 1*1+-1*2+4*0 & 1*2+-1*5+4*-3 & 1*0+-1*-3+4*6 \end{bmatrix} \begin{bmatrix} 2 & 1 & 6 \\ 6 & 18 & -15 \\ -1 & -15 & 27 \end{bmatrix}.$$

c. Compute BA. Is $AB = BA$ here? Was this guaranteed in general? [2 marks]

$$B*A = \begin{bmatrix} 1*2+2*0+0*1 & 1*0+2*3+0*-1 & 1*1+2*-1+0*4 \\ 2*2+5*0+-3*1 & 2*0+5*3+-3*-1 & 2*1+5*-1+-3*4 \\ 0*2+-3*0+6*1 & 0*0+-3*3+6*-1 & 0*1+-3*-1+6*4 \end{bmatrix} = \begin{bmatrix} 2 & 6 & -1 \\ 1 & 18 & -15 \\ 6 & -15 & 27 \end{bmatrix}$$

BA Equal AB. This is not guaranteed in general.

d. In R, verify parts (a)–(c) in R. Also compute `A * B` (element-wise) and explain in one sentence why this is a different operation from `A %*% B`. [4 marks]

```
A <- matrix(c(2,0,1,0,3,-1,1,-1,4), nrow = 3, byrow = TRUE)
B <- matrix(c(1,2,0,2,5,-3,0,-3,6), nrow = 3, byrow = TRUE)
A+B
3*A -2*B
A*B
A%*%B
B%*%A
```

AB is where is the elements are multiplied by the corresponding elements and that A%%B multiples the matrix properly

Q4

Let

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

a. Write down $X^\top$ and state its dimensions. [2 marks]

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

b. Compute $X^\top X$ and $XX^\top$ by hand. Explain why these two products have different dimensions even though both involve X and $X^\top$. [4 marks]

$$X^T X = \begin{bmatrix} 1*1+2*2+3*3 & 1*4+2*5+3*6 \\ 4*1+5*2+6*3 & 4*4+5*5+6*6 \end{bmatrix}. = \begin{bmatrix} 14 & 32 \\ 32 & 916 \end{bmatrix}. XX^T = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 214 & 16 \\ 27 & 16 & 45 \end{bmatrix}.$$

They have different dimensions because multiplication is Matrix1(ixj) x Martrix2 (jxi) in this case when you multiple Matrix1(ixj) x Martrix2 (jxi) you get matrix1x2(ixi) similarly if you reverse is Martrix2 (jxi) xMatrix1(ixj) matrix2x1(jxj)

c. Let $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{v}^\top\mathbf{v}$ and $\mathbf{v}\mathbf{v}^\top$ by hand, and state the dimension of each result. [4 marks]

$\mathbf{v}^\top\mathbf{v}$

$\mathbf{v}\mathbf{v}^\top = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \end{bmatrix}$

$\begin{bmatrix} 1*1+-1*-1+2*2 \end{bmatrix} = \begin{bmatrix} 6 \end{bmatrix}$

$\mathbf{v}\mathbf{v}^\top$

$\mathbf{v}^\top\mathbf{v} = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

$\begin{bmatrix} 1*1+-1*-1+2*2 \end{bmatrix} = \begin{bmatrix} 6 \end{bmatrix}$

d. In R, verify parts (a)–(c). Then confirm numerically, for two conformable matrices of your choosing, that $(AB)^\top = B^\top A^\top$. [4 marks]

```r
v <- matrix(c(1,-1,2), nrow = 3, byrow = TRUE)
v_transpose <- matrix(c(1,-1,2), nrow = 3, byrow = FALSE)
v*v_transpose
v_transpose * v
dim(v*v_transpose)
dim(v_transpose * v)
```

Q5

Consider

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}.$$

a. In R, compute A^{-1} using `solve()`. [2 marks]

```r
A <- matrix(c(4,2,3,3), nrow = 2, byrow = TRUE)
solve(A)
```

b. Construct a 3×3 matrix that is singular (for example, make one column a linear combination of the others) and show what happens when you call `solve()` on it. [2 marks]

```r
J <- matrix(c(8,4,2,4,2,1,1,1,1), nrow = 3, byrow = TRUE)
solve(J)
```

When I use solve I get an error message

Q6

a. Define, in your own words, what it means for a set of vectors to be linearly independent. [1 mark]

To be linear independent is where the vector cannot be linear combination of another vector. For example where a single vector cannot be a vector times by 2 of another vector. Also if you factor a multiple out of the vector the factored vector would not equal another.

b. Consider

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

Show that $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly dependent by finding constants c_1, c_2, c_3, not all zero, such that $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \mathbf{0}$. [3 marks]

```r
x1 <- c(1, 2, 1)
x2 <- c(0, 1, 1)
x3 <- c(2, 5, 3)

Z <- cbind(x1, x2, x3)
```

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \mathbf{0}\ c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} c_1 \\ 2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} c_1 & 0 & 2c_3 \\ 2c_1 & c_2 & 5c_3 \\ c_1 & c_2 & 3c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad c_1 + 2c_3 = 0$$

$2c_1 + c_2 + 5c_3 = 0$

$c_1 + c_2 + 3c_3 = 0$

$c_1 = -2c_3$ Then choose a value for c_3 $c_3 = 1$

$c_1 = -2(1) = -2$

$c_1 + c_2 + 3c_3 = 0$

$(-2) + c_2 + 3(1) = 0$

$c_2 = -1$

c. Let $M = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_3 \end{bmatrix}$. What is rank$(M)$? Explain your reasoning using your answer to part (b). [2 marks]

M would only have a rank of 2. Since the vectors are dependent on each other it is only a rank of 2. Because the second row is dependent on the first row. Such that the second row is equal to its own value plus 1.

d. In R, build M and confirm its rank using `qr(M)$rank`. Then modify one entry of $\mathbf{x}_3$ so that the three vectors become linearly independent, and confirm the new rank. [3 marks]

```
M <- matrix(c(1,0,2,2,1,5,1,1,3), nrow = 3, byrow=TRUE)
qr(M)$rank
M_2 <- matrix(c(1,0,7,3,10,5,1,1,3), nrow = 3, byrow=TRUE)
qr(M_2)$rank
```

Q7.

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

a. Write out the quadratic form $\mathbf{x}^\top A\mathbf{x}$ in terms of x_1 and x_2. [2 marks]

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \mathbf{X}^\top = \begin{bmatrix} x_1 & x_2 \end{bmatrix}, A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ $AX = \begin{bmatrix} 2 * x_1 + 1 * x_1 \\ 1 * x_2 + 2 * x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + x_1 \\ x_2 + 2x_2 \end{bmatrix}$ $XTA =$
$\begin{bmatrix} x_1 * 2x_1 + x_2 * x_2 \\ x_1 * x_1 + x_2 * 2x_2 \end{bmatrix} = \begin{bmatrix} 2x_1^2 + x_2^2 \\ x_1^2 + 2x_2^2 \end{bmatrix}$

b. Is B invertible? Justify your answer using the relationship between definiteness, rank, and invertibility discussed in class. [3 marks]

B is not invertible because it does not have full rank. Because the rows and columns are directly related to another it makes it that it does not have full rank

c. In R, Confirm parts (a) and (b) by computing `eigen(A)$values` and `eigen(B)$values`. State the rule connecting the sign of a symmetric matrix's eigenvalues to whether it is positive definite, positive semidefinite, or neither. [4 marks]

```r
A <- matrix(c(2,1,1,2), nrow = 2, byrow=TRUE)
B <- matrix(c(1,1,1,1), nrow = 2, byrow=TRUE)
eigen(A)$values
eigen(B)$values
```

Q8

Let

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

a. Show by direct calculation that $A^2 = A$, i.e., that A is idempotent. [2 marks]

$A^2 = \begin{bmatrix} 0.5^2 + 0.5^2 & 0.5^2 + 0.5^2 \\ 0.5^2 + 0.5^2 & 0.5^2 + 0.5^2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ Therefore A is Idempotenet

b. Using only the idempotent property $A^2 = A$ (not this specific matrix), show that $I - A$ is also idempotent. [2 marks]

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$I - A_1 = \begin{bmatrix} 1-1 & 0-0 \\ 0-0 & 1-0 \end{bmatrix} = \begin{bmatrix} 0 & 0-0 \\ 0-0 & 1 \end{bmatrix}$

$(I - A_1)^2 = \begin{bmatrix} 0^2 & 0^2 \\ 0^2 & 1^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

c. In R, confirm that A is idempotent using matrix multiplication in R. [1 mark]

```r
A <- matrix(c(0.5,0.5,0.5,0.5), nrow=2)
A%*%A
A_1 <- matrix(c(1,0,0,0), nrow=2)
A_1%*%A_1
```

Q9

Let $\mathbf{a} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and

$$S = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Note that S is symmetric.

a. Write down $\dfrac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}}$ for a general 3×1 vector $\mathbf{x}$. (No calculation needed, state the rule and apply it to this $\mathbf{a}$.) [2 marks]

general rule or derivation derivative in respect to X is 1 therefore the derivative would be a^T $\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = a^T = \begin{bmatrix} 3 & -2 & 1 \end{bmatrix}$

b. Derive $\dfrac{\partial(\mathbf{x}^\top S\mathbf{x})}{\partial \mathbf{x}}$ using the general rule for $\dfrac{\partial(\mathbf{x}^\top A\mathbf{x})}{\partial \mathbf{x}}$, and simplify using the fact that S is symmetric. [2 marks]

$\frac{\partial(x^T Ax)}{\partial \mathbf{x}} = (A + A^T)x$
$S = S^T$
$\frac{\partial(x^T Sx)}{\partial \mathbf{x}} = (S + S)x = 2Sx$
$2 \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

c. Evaluate your answer to part (b) at $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$. [1 mark]

$$\begin{bmatrix}2 & 1 & 0\\ 1 & 3 & 1\\ 0 & 1 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}4 & 2 & 0\\ 2 & 6 & 2\\ 0 & 2 & 4\end{bmatrix}\begin{bmatrix}1\\ 0\\ -1\end{bmatrix} = \begin{bmatrix}4*1+2*0+0*-1\\ 2*1+6*0+2*-1\\ 0*1+0*2+4*-1\end{bmatrix} = \begin{bmatrix}4\\ 0\\ -4\end{bmatrix}$$

d. In R, using the `numDeriv` package (`grad()`), numerically confirm your answers to parts (a) and (c). [2 marks] ::: callout-tip

```
install.packages("numDeriv")
library(numDeriv)
a <- matrix(c(3, -2, 1), nrow=3)
```

:::

Q10

Let $\mathbf{X} = \begin{bmatrix}X_1\\ X_2\\ X_3\end{bmatrix}$ be a random vector with

$$\mu = E[\mathbf{X}] = \begin{bmatrix}2\\ 0\\ -1\end{bmatrix}, \qquad \Sigma = \mathrm{Var}(\mathbf{X}) = \begin{bmatrix}4 & 1 & 0\\ 1 & 2 & -1\\ 0 & -1 & 3\end{bmatrix}.$$

Let $A = \begin{bmatrix}1 & 1 & 0\\ 0 & 1 & -1\end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix}5\\ -2\end{bmatrix}$, and define $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$.

a. State $E[\mathbf{Y}]$ and $\mathrm{Var}(\mathbf{Y})$, citing the properties used. [3 marks]

$$E[Y] = AE[X] + bVar[Y] = AVar[X]A^T$$

b. Compute $E[\mathbf{Y}]$ by hand. [2 marks]

$$
\begin{aligned}
E[Y] &= AE[X] + b \\
E[Y] &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \end{bmatrix} \\
&= \begin{bmatrix} 1*2+1*0+0*-1 \\ 0*2+1*0+-1*-1 \end{bmatrix} \\
&= \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \end{bmatrix} \\
&= \begin{bmatrix} 7 \\ -1 \end{bmatrix}
\end{aligned}
$$

c. Compute $\text{Var}(\mathbf{Y}) = A\Sigma A^\top$ by hand, showing your intermediate matrix products. [3 marks]

$$
\begin{aligned}
Var(Y) &= AVar(X)A^T \\
&= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 4*1+1+0 & 0+1+0 \\ 1+2+1 & 0+2+0 \\ 0+-1+-3 & 0+-1+0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 4 & 2 \\ -4 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 1*4+4+0 & 1+2+0 \\ 0+4+4 & 0+2-1 \end{bmatrix} \\
&= \begin{bmatrix} 8 & 3 \\ 8 & 1 \end{bmatrix}
\end{aligned}
$$

d. Let $\mathbf{a} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\text{Var}(\mathbf{a}^\top \mathbf{X})$ using the appropriate property, without recomputing $\text{Var}(\mathbf{X})$ from scratch. [3 marks]

$$Var(a^T X) = a^t Var(X) a$$

e. In R, using `MASS::mvrnorm()`, draw 100,000 observations from a distribution with mean μ and variance Σ (you may treat $\mathbf{X}$ as multivariate normal for this simulation). Confirm your answers to (b), (c), and (d) using `colMeans()` and `cov()`. [3 marks]

```r
library(MASS)

mu <- c(2, 0, -1)

Sigma <- matrix(c(
  4, 1, 0,
  1, 2, -1,
  0, -1, 3
), nrow = 3, byrow = TRUE)

A <- matrix(c(
  1, 1, 0,
  0, 1, -1
), nrow = 2, byrow = TRUE)

b <- c(5, -2)

a <- c(1, -1, 2)

Xsim <- mvrnorm(100000, mu, Sigma)

Ysim <- Xsim %*% t(A) + matrix(b, nrow = 100000, ncol = 2, byrow = TRUE)

colMeans(Ysim)
cov(Ysim)

asim <- Xsim %*% a

var(asim)
```

Q11

a. Suppose $\mathbf{X} \sim N_3(\mu, \Sigma)$ using the μ and Σ from Question 10. What is the marginal distribution of X_2 alone? Give its distribution and parameters. [2 marks]

Distribution of X_2 $N(0, 2)$

c. Using property 4 of the multivariate normal (zero covariance implies independence), determine whether X_1 and X_3 are independent. Justify your answer using an entry of Σ. [1 mark]

$\sum = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ $Cov(X_1, X_3) = 0$ We find that they are independent

d. In R, using your simulated draws from Question 10(e), produce a scatterplot of X_1 against X_3 and compute `cor(X1, X3)`. Comment on whether the plot is consistent with your answer to part (c). [3 marks]

```
plot(Xsim[,1], Xsim[,3],
     xlab = "X1",
     ylab = "X3",
     main = "Scatterplot of X1 against X3")
```